

Chain-of-Conceptual-Thought: Eliciting the Agent to Deeply Think within the Response

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Abstract

Chain-of-Thought (CoT) is widely applied to enhance the LLM capability in math, coding and reasoning tasks. However, its performance is limited for open-domain tasks, when there are no clearly defined reasoning steps or logical transitions. To mitigate such challenges, we propose a new prompt-based paradigm called Chain of Conceptual Thoughts (CoCT), which suggests the LLM first to produce the tag of concepts, then complete the detailed content following the concept. To encourage this hierarchical way of thinking, we implement the concepts with emotions, strategies and topics. We experiment with this paradigm in daily and emotional support conversations, covering tasks with both in-domain and out-of-domain concept settings. Automatic, human, and LLM-based evaluations reveal that CoCT surpasses several prompt-based baselines such as self-refine, ECoT, SoT and RAG, suggesting a potential solution of LLM prompting paradigm for a wider scope of tasks.

1 Introduction

Large Language Models (LLMs) have demonstrated remarkable capabilities in tasks of natural language understanding, question-answering, daily conversation, and complicated reasoning (Ouyang et al., 2022; AI@Meta, 2024). To further elicit their reasoning abilities, a Chain-of-Thought (CoT) paradigm (Wei et al., 2022) is typically adopted, which prompts the LLM to think ‘step by step’ before producing the final response. Several effective paradigms are then proposed by incorporating CoT with specific reasoning hierarchies such as Tree of Thoughts (Yao et al., 2023a), Graph of Thoughts (Besta et al., 2024), Structured Chain-of-Thought (Li et al., 2025), or with test-time scaling (Yao et al., 2023b; Shinn et al., 2023; Madaan et al., 2023; Ning et al., 2024).

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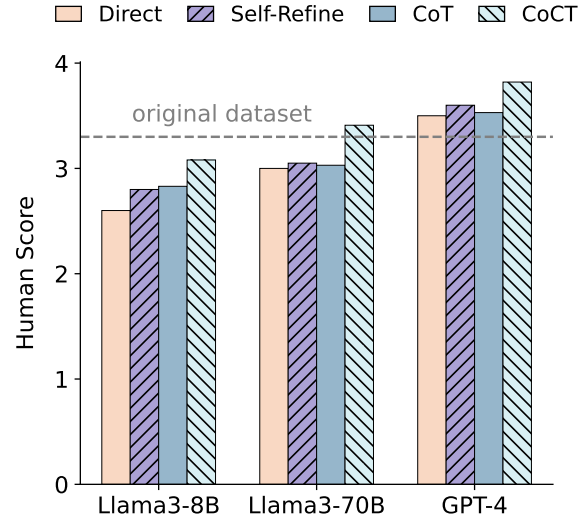


Figure 1: Comparison between CoCT and other prompt-based baselines on ESconv.

Although effective, such CoT-related methods are often focused on math problems, code generation, or other reasoning scenarios that have clear reasoning steps. When dealing with open-domain tasks such as conversation, CoT (or CoT-like methods on decision trees or graphs) (Wei et al., 2022; Yao et al., 2023a; Besta et al., 2024; Li et al., 2025) may encounter difficulties since conversations generally lack clear definitions of reasoning steps, step transitions, and stepwise standardized answers. To alleviate such issues, ECoT (Li et al., 2024; Kang et al., 2024) is proposed which encourages sequential emotional decisions on emotional support conversations (ESC). SoT (Ning et al., 2024) instead first determines the skeleton tokens, then generates the completed response. However, methods such as ECoT and SoT still do not provide solutions to open-domain conversations and also introduce excessive computational costs from iterative inference.

To bridge this gap, we aim to propose an alternative, generalized prompting paradigm that en-

Table 1: Lists of annotated emotions, strategies and topics from ESConv (Liu et al., 2021), DailyDialogue (Li et al., 2017), EmpatheticDialogues (Rashkin et al., 2019) and Cskills (Zhou et al., 2024), respectively. Strategies of ESConv are abbreviated (for full names, refer to (Liu et al., 2021)). Only the most frequent 10 emotions of EmpatheticDialogues and the most frequent 8 topics of Cskills are listed.

Dataset	Concepts	Concept Categories
ESConv	<i>Emotion</i>	anger, anxiety, depression, disgust, fear, nervousness, sadness, shame
	<i>Strategy</i>	Que., Paraphrasing & Res., Ref., Self-Dis., Aff. & Rea., Pro., Inf., Others
DailyDialogue	<i>Emotion</i>	anger, disgust, fear, happiness, sadness, surprise, no emotion
	<i>Strategy</i>	inform, question, directive, and commissive
EmpatheticDialogues	<i>Emotion</i>	surprised, grateful, proud, sentimental, annoyed, excited, sad, disgusted, . . .
Cskills	<i>Topic</i>	sports, travel, art, music, technology, food&drink, hobbies, entertainment, . . .

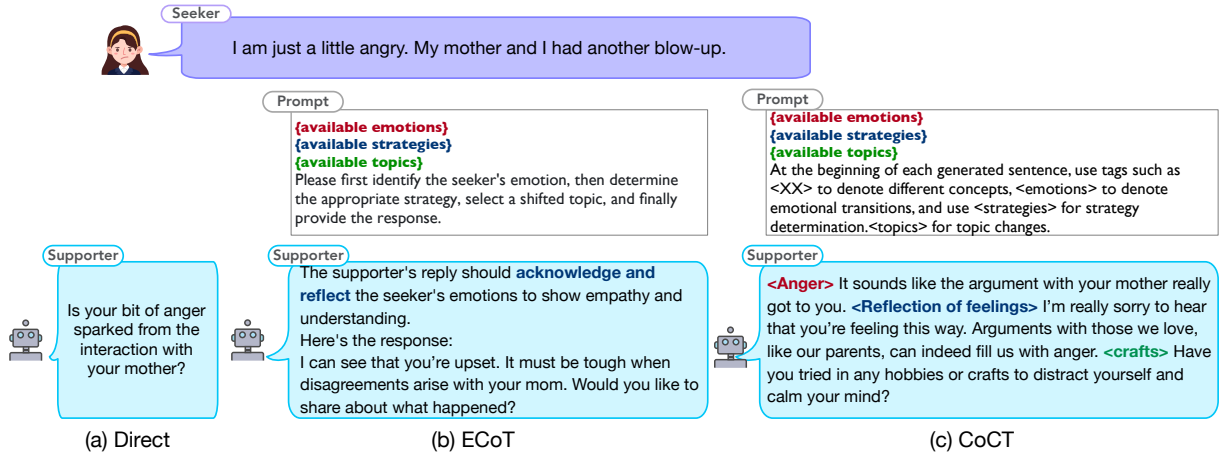


Figure 2: The paradigm of CoCT. The red bold text represents emotions, the blue bold text represents strategies, and the green bold text represents topics of conversation transitions.

hances LLMs in broader, open domains, such as daily and emotional support conversations. Our idea is motivated by the hierarchical nature of human thinking, where a person generally first formulates a list (or chain) of strategic concepts, each of which may then guide detailed thoughts (team et al., 2024; Ning et al., 2024). This chain of concepts and corresponding concept-guided contents formulates the ultimate human thought. For example, a customer may deliberately first exaggerate the product’s disadvantages, then make a solid decision during a bargain; a psychological counselor may first express an empathetic attitude, then provide a detailed, practical solution. In daily life, we summarize these concepts into several primary types, including:

- **Emotion:** the agent’s emotional or sentiment states, such as joy, sadness, or anger.
- **Strategy:** the agent’s conversational strategy, such as asking for more details or summariz-

ing the contents.

- **Topic:** the agent might choose to deepen the topic or shift the user’s intention to a new topic for better effects.

Table 1 exemplifies these concept terms from the annotations of previous textual datasets (Li et al., 2017; Rashkin et al., 2019; Zhou et al., 2024). We argue that LLM can utilize these concept annotations in the new prompt paradigm, resulting in a new hierarchical way of thinking.

In this paper, we propose a straightforward but effective paradigm called **Chain of Conceptual Thought (CoCT)**, which introduces a *conceptual chained* variant of the classical CoT, as the solution to open-domain or daily conversations. For each response utterance, we prompt the LLM to first propose a concept tag (might be one instance of either emotion, strategy, or topic, surrounded by special tokens), then generate the content grounded by that concept. Figure 1 provides a quick case comparison

between Direct inference, ECoT and CoCT. CoCT showcases in both open-domain and emotional support conversation tasks, either given a pre-defined, fixed concept list (refer to Table 1), or based on dynamic concept generation. CoCT also generalizes well to out-of-domain concepts and queries. Based on automatic, human and LLM-based evaluations, CoCT outperforms other prompt-based baselines, including Direct inference, Self-Refine, ECoT, SoT and RAG. Figure 1 provides a quick snapshot of the human scoring comparison. To summarize, our key contributions include:

- We propose CoCT, a prompt-based paradigm which allows the LLM to think conceptually and strategically in daily conversation or open-domain tasks.
- We test CoCT on tasks of daily conversation and emotional support conversations, and obtain better response quality than strong baselines.
- CoCT is robust to out-of-domain concepts and queries, and can also perform reasonably with concepts generated on-the-fly.

2 Method

This section introduces our method details, including the prompt paradigm, the possible concepts, and resulting inner-utterance concept transitions.

2.1 Chain-of-Conceptual-Thought Prompt

According to the studied scenario, we assume a pre-defined list of concepts $\{c\}$ is available based on prior knowledge. These concepts are then applied to steer the LLM generation

$$output \leftarrow \text{LLM}(\{c_1, c_1, \dots, c_n\}, \mathcal{I}, input) \quad (1)$$

in which $\mathcal{I}$ is the instruction template of CoCT, c is a concept and n is the total number of concepts.

Below is the template of our CoCT prompt:

CoCT Prompt Format

‘{possible concepts}’,
At the beginning of each generated response, use tags such as <XX> to denote different concepts, then follow the sentence content.
Use chain of concepts to denote the concept transitions.

To summarize, the CoCT prompt $\mathcal{I}$ starts from a detailed definitions of the conceptual list $\{c\}$, which helps the LLM extract usable concepts from potentially unlimited space. After that, for each generated sentence, we ask the LLM to first decode the special token of concept, then decode the detailed sentence content.

2.2 Formulation of Concepts

As discussed in Section 1, in this paper we study three main concept types: **emotion**, **strategy** and **topic**. While we find CoCT can also work with concepts generated on-the-fly, better performance can be obtained with *pre-defined* concept lists. As indicated by Table 1, we get the concepts of **emotion** and **strategy** directly from the annotations of ESConv and DailyDialogue. For **topic**, we refer to the topic guidance study in Cskills, which provides about 8 topics after frequency filtering: *sports, travel, art, music, technology, food and drink, hobbies and crafts, entertainment and animal*.

2.3 CoCT Elicits the Inner-utterance Transitions

Human talk sometimes implies the transition of concepts within a single utterance; however, nominal LLM-based conversation usually fails to do so. Instead, one can observe that each of their responses contains a single emotion or topic, and such concepts’ transitions happen on the outer-utterance (Figure 3 (Left)). On the other hand, CoCT naturally encourages the inner-utterance concept transitions (Figure 3 (Right)), which is part of our main purpose:

$$c_i \rightarrow c_j, \quad i, j \in \{1, \dots, n\} \quad (2)$$

3 Experiment

In this section, we first briefly introduce the experimental setting, including the implementation, baseline and metric details. Then we exhibit the effect of CoCT on daily conversations. To provide more quantitative results, we then conduct more experiments on emotional support conversations. Finally, we provide some insightful analysis.

3.1 Setting

Implementation. The experiment is conducted on vLLM, with 8 A100 80GB PCIe GPUs. The context window is restricted to 4096. To validate the effectiveness of CoCT on different model scales, we

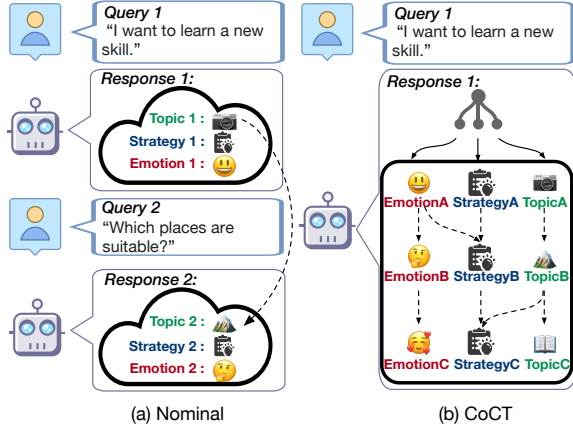


Figure 3: Comparative visualizations between outer-utterance transitions (Left) and inner-utterance transitions (Right). The dash arrows indicate the conceptual transitions between emotions, strategies or topics.

experiment with the performances on Llama3-8B-instruct (AI@Meta, 2024), Llama3-70B-instruct (AI@Meta, 2024) and GPT4o, respectively.

Datasets and tasks. In this paper, we conduct CoCT on both in-domain (ID) and out-of-domain (OOD) tasks. In ID tasks, the datasets have annotations of emotions, strategies, or topics, which are used as the possible concepts $\{c\}$ during the test. For OOD tasks, the datasets do not have such concept annotations. Therefore, we use $\{c\}$ defined in ID tasks, and test CoCT’s generalization capability of these concepts to this new dataset. Since it is more common to assume there are no conceptual annotations, the OOD tasks can reflect the capability of CoCT in most scenarios. The conversational datasets we use can be classified into two main categories:

- (1) Open-ended conversations: we use the annotated lists of emotions and strategies from *DailyDialogue* (Li et al., 2017), and the list of topics from *Cskills* (Zhou et al., 2024). Together, we have 7 emotions, 4 strategies and about 20 topics. Correspondingly, we conduct the ID test of CoCT on the test set of *DailyDialogue* and *Cskills*. After that, we apply CoCT on another open-topic dataset *MultiWoZ* (Budzianowski et al., 2018) as the OOD test, which does not have concept annotations.
- (2) Emotional support conversations: we then investigate the performance of CoCT on this specific scenario for more in-depth analysis. We employ 11 emotions and 8 strategies from *ESConv*, conduct the ID test of CoCT on the test of *ESConv*; and the OOD test on *EmpatheticDialogues* (Rashkin et al.,

2019).

Appendix A provides more dataset details.

Metrics. Besides the cases, the response quality can also be evaluated by human annotations, automatic metrics and LLM. Here we briefly introduce the details.

- **Automatic Metrics:** Include similarity-based metrics such as BLEU-2 (**B-2**) (Papineni et al., 2002), Rouge-L (**R-L**) (Lin, 2004) and CIDEr (**CDr**) (Vedantam et al., 2015); and diversity-based metric Distinct-2 (**D-2**).
- **LLM-Based Simulations:** To qualitatively evaluate the quality of daily conversation, we utilize the Cskills benchmark (Zhou et al., 2024), with about 1000 queries. The agent is expected to generate reasonable and qualified dialogue content. Simulation stops when there is no significant intention to communicate. The averaged utterance length and dialogue rounds are employed as metrics.
- **LLM-as-a-Judge:** We used GPT-4o to compare different responses and provide the win-tie-lose suggestion. The evaluation prompt is from (Madani et al., 2024).
- **Human Scoring:** four annotators are employed to provide quality scoring. The cross-validation is conducted with the averaged results shown later. Similar to (Kang et al., 2024), the scoring dimensions include *Acceptance*, *Effectiveness*, *Sensitivity*, *Fluency*, and *Emotion*, and the *user’s Satisfaction*.

We provide more evaluation details in Appendix B, such as computation formula, annotation principle and evaluation prompts.

Baselines. As fair comparisons, we implement several prompt-based baselines, with the same base model and context. Such baselines can be classified into several categories: (1) *Direct* inference the LLM, *Direct-Refine* the result within the same response, and Self-Refine (Madaan et al., 2023) the result by inference the second time. (2) Methods based on chained or other structured data format, including *ECOT* (Li et al., 2024), *SoT* (Ning et al., 2024), *ToT* (Yao et al., 2023a), and *Plan-and-Solve* (Wang et al., 2023). (3) Retrieval augmented generation (RAG) (Fan et al., 2024) which retrieves the suitable concepts and descriptions and prompts

the LLM to respond. In this work, we employ E5-Large (Wang et al., 2024) as the retriever. (4) *CSIM* (Zhou et al., 2024): a self-chat method which produces the response based on its implicit simulation of future dialogue.

More details can be found in Appendix C.

3.2 Results on Daily Conversation

We first test CoCT’s performance on daily conversations, with no explicit constraints on topics.

Performance on in-domain concepts. Table 2 shows the automatic metrics of DailyDialogue, where CoCT performs the best with the basis on LLaMA3-8B-Instruct and GPT-4o, indicating it can generate responses in a similar style to the original annotated response. CoCT performs the second-best on LLaMA3-70B-Instruct (worse than Direct-Refine), which we suppose may be due to insufficient exploration when we apply the prompt tested on 8B directly to 70B.

Generalize with out-of-domain concepts or queries. Table 2 also exhibits the result of MultiWOZ on LLaMA3-8B-Instruct and LLaMA3-70B-Instruct. CoCT still performs the best on B-2, R-L and CDr, indicating its strong generalization capabilities on different topics and domains. On the other hand, CoCT is the second-best on D-2, a diversity-based metric, which is reasonable since the pre-defined concepts may constrain the methodology’s scope¹.

Cases on more situations To further verify the generalization capability of CoCT, in this experiment we test it with arbitrary user queries. To explore more situations, we experiment with three combinations of concept types: with strategy only (w/ strategy), with emotion only (w/ emotion), or with both emotion and strategy (w/ emotion+strategy). Furthermore, we also attempt to expand the strategies with two more arbitrary tags: `<Humor>` and `<Interest>`. We expect such new tags can be automatically recognized by LLM and provide more interesting responses.

Table 4 showcases CoCT’s performance on these arbitrary scenarios. It can be observed that CoCT still performs well on these out-of-domain queries and concepts, generating reasonable concept transitions. Especially, `<Humor>` is utilized in the first case and `<Interest>` is utilized in the third case,

generating more funny responses and steering the conversation to a higher communication level.

3.3 Results of Emotional Support Conversations

In this subsection, we further investigate the capability of CoCT on a specific domain of conversations, the emotional support conversation (ESC).

Automatic Metrics. The last two columns of Table 2 illustrate the ID results on ESConv and the OOD results on EmpatheticDialogues. Similar to tests on daily conversations, CoCT is still the best on B-2, R-L and CDr. This observation indicates CoCT holds stable and robust performance for this specific domain.

Human Scores. Table 5 shows the human annotation results. CoCT has higher scores than direct inference, direct-refine, self-refine and CoT, with the bases of Llama3-8B-Instruct, Llama3-70B-Instruct and GPT-4. Figure 1 in Section 1 provides a snapshot of the Satisfaction scores. It is also worth mentioning that although the original dataset is originally annotated as expert demonstrations, CoCT based on Llama3-70B-Instruct starts to surpass the original dataset, while other baselines can not. For results based on GPT-4o, the human scores are all higher than the original dataset, due to the stronger model basis. Our CoCT still remains the best.

Ablation Study. We conduct several ablations, including *w/o strategy* and *w/o emotion* in the concept list. We compare their performances to the formal CoCT on human and automatic evaluations, as well as the pairwise evaluation (versus original dataset) by GPT-4o. Table 6 shows that CoCT still outperforms these ablations, indicating both strategy and emotion are helpful concept types.

3.4 Discussion

Conceptual transition within utterance. Compared to previous approaches, CoCT allows for the transitions between different concepts (e.g., recognition of emotions and strategies) within a single utterance, which is aligned with the human-like thinking behavior, and closer to the original annotated response of ESConv. To further illustrate this behavior, we exhibit the distribution of the inner-utterance conceptual transitions in Figure 4. For a specific grid (i, j) , i denotes the row index and j

¹We will discuss the improvement method in Section 3.4.

Table 2: Results on automatic metrics on daily conversations (DailyDialog and MultiWoZ) and ESC (ESConv and EmpatheticDialogues). E.D. abbreviates EmpatheticDialogues. The best results on each LLM are **bolded** and the second best are underlined.

Method ↓	DailyDialogue (ID)			MultiWoZ (OOD)				ESConv (ID)			EmpatheticDialogues (OOD)			
	B-2	R-L	CDr	B-2	R-L	D-2	CDr	B-2	R-L	CDr	B-2	R-L	D-2	CDr
<i>LLaMA3-8B-Instruct</i>														
+ Direct	<u>3.35</u>	<u>10.33</u>	1.99	<u>4.94</u>	<u>13.18</u>	16.09	4.12	<u>3.47</u>	<u>10.64</u>	1.14	<u>3.09</u>	<u>9.91</u>	25.23	1.60
+ Direct-Refine	2.56	8.70	1.75	4.02	11.90	17.92	1.57	3.10	6.13	1.30	2.56	9.12	22.32	0.42
+ Self-Refine	2.40	7.75	1.27	4.47	12.61	19.13	3.07	3.34	9.71	<u>1.79</u>	3.08	<u>9.91</u>	25.20	<u>1.56</u>
+ ECoT	1.78	6.00	<u>2.43</u>	4.22	11.89	26.28	2.36	3.16	10.50	1.50	2.91	9.79	<u>32.65</u>	1.37
+ Plan-and-Solve	2.60	7.76	1.92	4.32	10.74	18.99	<u>4.87</u>	2.81	8.27	1.59	2.69	6.93	24.02	0.99
+ ToT	2.52	8.84	0.61	3.06	10.05	15.98	0.23	2.65	9.81	0.60	2.31	9.05	29.09	0.12
+ SoT	2.53	7.97	1.76	3.58	9.77	17.09	3.55	3.07	8.76	1.70	1.79	5.66	48.59	0.19
+ CoCT (ours)	3.62	11.64	3.84	6.94	17.95	<u>19.79</u>	9.73	4.75	13.31	4.87	3.29	10.84	26.27	1.08
<i>LLaMA3-70B-Instruct</i>														
+ Direct	3.50	11.20	2.46	<u>5.19</u>	<u>14.38</u>	14.38	1.37	3.06	10.16	0.8	2.60	8.90	26.71	0.44
+ Direct-Refine	4.16	12.84	6.04	4.63	13.32	15.65	1.40	2.54	9.40	0.27	2.53	9.07	25.28	0.30
+ Self-Refine	3.40	10.83	2.97	4.82	13.47	17.09	1.51	2.97	10.12	1.07	<u>2.81</u>	<u>9.74</u>	27.06	0.48
+ ECoT	2.09	7.62	1.23	4.82	13.47	17.09	1.51	1.85	9.67	1.03	2.60	9.06	34.02	<u>1.14</u>
+ Plan-and-Solve	2.13	6.32	0.80	3.96	10.41	13.49	3.09	2.01	8.68	1.36	2.14	6.25	37.16	0.50
+ ToT	2.68	9.43	0.69	4.08	13.91	16.62	0.40	2.39	8.96	0.16	2.16	8.24	27.14	0.13
+ SoT	2.72	8.55	1.72	3.92	11.01	20.80	<u>3.96</u>	3.13	9.49	1.16	2.12	6.83	45.98	0.35
+ RAG	2.78	9.67	1.67	-	-	-	-	<u>3.44</u>	<u>10.95</u>	<u>1.65</u>	2.10	8.18	28.92	0.13
+ CoCT (ours)	<u>3.87</u>	<u>11.94</u>	<u>3.81</u>	6.45	17.00	<u>17.34</u>	4.78	4.30	12.71	2.07	3.48	11.20	<u>29.34</u>	1.45
<i>GPT-4o</i>														
+ Direct	<u>4.25</u>	<u>12.80</u>	6.09	-	-	-	-	3.48	<u>13.21</u>	<u>3.83</u>	<u>3.24</u>	<u>10.70</u>	<u>36.64</u>	1.54
+ Direct-Refine	4.03	12.22	4.49	-	-	-	-	4.19	12.61	2.65	-	-	-	-
+ Self-Refine	3.19	9.84	3.19	-	-	-	-	5.40	12.70	2.82	-	-	-	-
+ ECoT	3.48	11.10	3.46	-	-	-	-	3.64	11.64	1.95	2.30	7.84	26.87	0.07
+ RAG	4.15	9.75	<u>7.14</u>	-	-	-	-	3.68	11.05	3.67	2.60	8.63	53.66	0.95
+ CoCT (ours)	4.60	13.25	8.73	-	-	-	-	<u>4.62</u>	13.77	4.79	3.47	11.20	32.36	<u>1.30</u>

Table 3: Results on the Cskills. We use the ‘concept guidance’ subset of the original benchmark.

Method →	Direct	CSIM	RAG	CoCT (ours)
AvgLen	46.41	47.73	49.13	67.04
Rounds	3.53	4.26	4.36	4.62

denotes the column index, and its color level indicates the relative occurrence of transition $c_i \rightarrow c_j$.

Figure 4 (left) exhibits the strategy transitions generated by CoCT on ESConv. According to (?), the strategies can be related to three supporting stages (Exploration, Comforting and Action), therefore we mark each strategy with the corresponding stage I, II and III and order the strategies accordingly. As a result, a more reasonable transition may happen from a left strategy to a right one, *i.e.*, the upper-triangle part of the matrix. Obviously, such a pattern can be observed from Figure 4 (left), indicating that the strategy determination of CoCT is aligned with the ‘Exploration, Comforting and

Action’ paradigm.

Figure 4 (right) shows the emotion transitions generated by CoCT on DailyDialog. From the distribution, we can observe that more frequent emotion transitions of CoCT happen on Disgust&Anger $\rightarrow$ Happiness, Anger $\rightarrow$ Disgust, and Fear&Sadness $\rightarrow$ Surprise. On the contrary, the over-extreme emotional transition (Happiness $\rightarrow$ Sadness) has a low frequency. CoCT mimics the human emotion curve when expressing opinions, providing a more fine-grained emotional interaction.

CoCT without pre-defined concepts. Previous results (especially OOD) suggest that the pre-defined concepts may hinder CoCT from exploring broader scopes. In Table 6, we also try to allow the LLM to generate concepts on-the-fly (*free concept*). Surprisingly, although *free concept* endures some performance degradation, it still performs well and surpasses the other baselines. This observation suggests the potential of CoCT as a more

Table 4: Open multi-turn cases generated by CoCT.

w/ strategy	user	<i>I've started learning how to cook.</i>
	CoCT	<Affirmation and Reassurance> That's fantastic! Learning to cook is such a valuable skill and can be really rewarding. <Question> What type of dishes are you interested in trying out?
	user	<i>I tried making pizza last week. It was a bit of a fail the first time, but it still tasted okay!</i>
	CoCT	<Reflection of feelings> It sounds like you were a bit disappointed with the first attempt, but also glad that it still turned out tasty. <Humor> Hey, even a "fail" pizza is still pizza, right? That's a win in my book! <Question> What part of making the pizza did you find the most challenging?
w/ emotion	user	<i>I've started learning how to cook.</i>
	CoCT	<Excitement> That's fantastic! Learning to cook is such a rewarding skill. What dish are you most excited to try making?
	user	<i>I tried making pizza last week. It was a bit of a fail the first time, but it still tasted okay!</i>
	CoCT	<Nervousness> First attempts can be tricky, but it's great that it still tasted okay! <Encouragement> Practice makes perfect, and every time you cook, you'll get better. What part of making the pizza did you find most challenging?
w/ emotion+strategy	user	<i>I've started learning how to cook.</i>
	CoCT	<Interest> That's fantastic! Cooking can be a rewarding and enjoyable skill to develop. <Questioning> What kinds of dishes are you learning to make?
	user	<i>I tried making pizza last week. It was a bit of a fail the first time, but it still tasted okay!</i>
	CoCT	<Affirmation> It's great that you tried making pizza! <Reassurance> Everyone has a few mishaps when they're learning something new. <Humor> Besides, an "okay" tasting pizza is still pizza, right? <Questioning> What part of the process do you think was the most challenging?

Table 5: Human evaluation of response quality on ESConv and EmpatheticDialogues. The best results of each LLMs are **bolded** and the second best are underlined.

Size	Method	Human Annotation						
		Fluency	Emotion	Acceptance	Effectiveness	Sensitivity	Alignment	Satisfaction
-	original dataset	3.51	3.61	3.40	3.10	3.50	3.20	3.30
8B	LLaMA3 (8B)	2.95	3.00	2.60	2.40	2.70	2.70	2.60
	+ Direct-Refine	3.09	3.09	2.73	2.91	<u>2.91</u>	2.82	<u>2.84</u>
	+ Self-Refine	3.10	<u>3.15</u>	2.80	2.70	2.90	2.80	2.80
	+ ECoT	<u>3.08</u>	3.08	<u>2.83</u>	2.67	<u>3.00</u>	<u>2.83</u>	2.83
	+ CoCT (ours)	3.10	3.25	3.00	3.20	3.10	3.00	3.08
70B	LLaMA3 (70B)	3.05	3.30	2.70	3.00	3.20	3.10	3.00
	+ Direct-Refine	3.40	<u>3.50</u>	<u>3.00</u>	<u>3.10</u>	<u>3.10</u>	2.90	3.03
	+ Self-Refine	3.40	3.35	2.90	3.10	3.00	3.20	3.05
	+ ECoT	3.25	3.30	2.80	3.00	3.10	3.20	3.03
	+ CoCT (ours)	<u>3.32</u>	3.77	3.36	3.55	3.55	<u>3.18</u>	3.41
larger	GPT-4o	3.56	3.67	3.67	3.56	3.56	3.22	3.50
	+ Direct-Refine	<u>3.80</u>	3.70	<u>3.60</u>	3.50	3.70	<u>3.60</u>	<u>3.60</u>
	+ Self-Refine	<u>3.80</u>	<u>3.75</u>	3.50	<u>3.70</u>	<u>3.60</u>	<u>3.60</u>	<u>3.60</u>
	+ ECoT	3.75	3.70	3.50	3.50	<u>3.60</u>	3.50	3.53
	+ CoCT (ours)	3.85	4.00	<u>3.60</u>	4.10	3.70	3.70	3.82

generalized methodology, *e.g.*, remove the concept list in the prompt, but pretrain the LLM with some conceptual knowledge.

CoCT with finetuning. Although CoCT is primarily proposed as a prompt-based paradigm, it

is also possible to collect the CoCT result from a stronger LLM, then finetune it on a smaller LLM. This methodology is also known as distillation from the teacher LLM. In Table 7, we compare the 70B result of CoCT with finetuning to the original result, indicating that CoCT is friendly to finetuning,

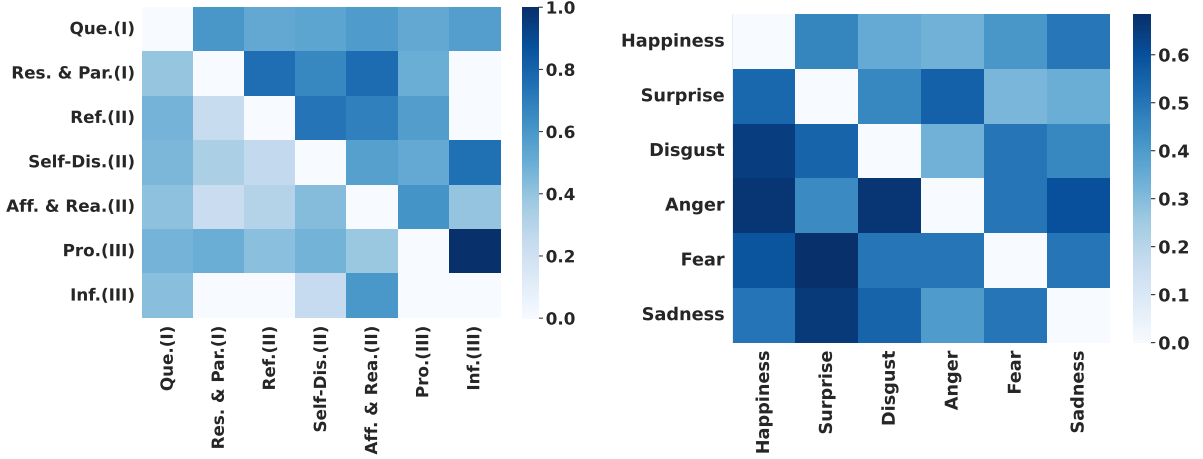


Figure 4: Distribution of conceptual transitions within the utterance. Left: the strategy transition on ESConv. Right: the emotion transition on DailyDialogue.

Table 6: Ablation study of CoCT on Llama3-8B-Instruct, with human, automatic and win-tie-lose rates (%) of GPT-4o evaluation results. Experiments are conducted on ESConv. ‘Satis.’ abbreviates Satisfaction. The best result is **bolded** and the second best is underlined.

Method ↓	human	automatic		VS dataset	
	Satis.	B-2	R-L	win ↑	lose ↓
wo/ strategy	2.80	4.25	12.67	51.90	37.40
wo/ emotion	3.00	4.03	12.60	60.90	33.70
free concept	2.90	<u>4.46</u>	<u>13.14</u>	66.86	16.75
CoCT	3.08	4.60	13.34	71.71	26.80

Table 7: The finetuning result of CoCT on LLaMA3-70B-Instruct, from responses of GPT4o.

Method ↓	ESConv			
	B-2	R-L	D-2	CDr
CoCT	4.30	12.71	18.16	2.07
CoCTw/ finetuning	4.75	13.49	42.54	8.68

and its performance can be further enhanced by distillation from a teacher, *i.e.*, GPT4o.

Choice of special tokens. The prompt of CoCT requires an arbitrary choice of the special token to tag the concept term. In Section , we take ‘<>’ as the example, while here we compare its results to other special tokens (e.g., #, @, [, &). Table 8 exhibits this comparison result on 8B and 79B, showing that the current choice is optimal on 70B, while different special tokens may perform similarly on 8B. In practice, one may conduct similar tests to determine the special token before the formal experiment.

Table 8: Performance comparison of different special tokens on ESConv.

Method ↓	ESConv			
	B-2	R-L	D-2	CDr
LLaMA3 (8B)				
w/ special token: <>	4.75	13.31	18.54	4.87
w/ special token: ^	4.38	13.16	18.34	3.79
w/ special token: #	4.86	<u>14.12</u>	18.65	4.99
w/ special token: @	4.54	13.54	18.43	4.07
w/ special token: []	<u>4.82</u>	14.13	<u>18.78</u>	5.09
w/ special token: &	4.72	13.71	19.04	4.62
LLaMA3 (70B)				
w/ special token: <>	4.30	12.71	18.16	2.07
w/ special token: ^	3.96	12.14	16.60	1.32
w/ special token: #	<u>4.17</u>	<u>12.68</u>	<u>17.04</u>	<u>1.88</u>
w/ special token: @	4.11	12.54	16.68	1.74
w/ special token: []	4.01	12.45	16.18	1.48
w/ special token: &	3.92	12.06	15.94	1.25

4 Related Work

Large language models can generalize to different tasks given an informative and instructional prompt. Chain-of-thought (CoT) (Wei et al., 2022) is a famous study that prompts the LLM by the paraphrase ‘Let’s think step by step’ and obtains substantial performance improvement on metathetical and reasoning tasks. Further studies are proposed to enhance the performance by using different prompt paradigms. For example, Chain-of-Hindsight (CoH) (Liu et al., 2024) allows the LLM to reflect from human feedback and have a better alignment with human preference. Self-Refine (Madaan et al., 2023) let the LLM generate feedback from its initial response, then refine the re-

sponse again based on this feedback. SoT (Ning et al., 2024) first guides LLMs to generate the skeleton of the answer, and then conducts parallel API calls or batched decoding to complete the contents.

In contrast, here we mainly focus on the open-ended conversation scenario and argue that LLM can also benefit from a chain of concepts, instead of reasoning steps. Such concepts may include emotions, strategies and topics, and can transit within the utterance. LLM is prompted to first generate the sketchpad of concepts, followed by the formal response.

5 Conclusion

In this work, we propose a novel prompt paradigm called CoCT which can be applied to open-domain conversations. Motivated by humans’ conceptual ideas such as emotions, strategies and topics, we let the LLM first generate the tag of the concept, then produce the detailed content of the corresponding concept. This chain of conceptual thoughts facilitates the LLM to have a hierarchical and strategic way of thinking, highlighting its decision transitions within a single utterance. Experiments on daily and emotional support conversations show that CoCT produces better responses than other prompt-based baselines, given either in-domain and out-of-domain concepts and queries. Instead of providing a pre-defined concept list, CoCT can also generate the answer with concepts generated on-the-fly, potentially implying a more generalized generative paradigm.

6 Limitation

CoCT is orthogonal to few-shot demonstrations, finetuning, self-reflection, and usage of structured or external knowledge. Combinations of these methods might further benefit.

Currently, CoCT also needs a manually provided list of concept candidates, which might constrain its application. Future work may consider invoking the LLM’s hierarchical thinking mechanism, which automatically first formulates the next key concept, then deduces the detailed response.

Although typical cases and quantitative metrics primarily indicate the effectiveness of CoCT, due to the capability limitation of the foundation model, CoCT still sometimes fails to understand the concept list, or conduct the semantically wrong conceptual transition. We also include a bad case analysis in the Appendix.

7 Ethical Considerations

CoCT can exhibit inconvenient, aggressive behaviors given specific concepts. It is suggested that the concept list needs to be carefully checked before the formal deployment. If concepts are generated on-the-fly, the finetuning may be needed to constrain the behavior within safety.

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A More Dataset Details

Detailed introductions. We introduce our experimental datasets in more detail;

- The *DailyDialogue* dataset (Li et al., 2017) is a widely used benchmark for Open-ended conversation. It contains 13,118 multi-turn dialogues across diverse daily topics (e.g., family, work, hobbies) with manually annotated emotions and dialogue strategies (named ‘act’ in the original dataset). It is annotated with 7 emotions and 4 strategies.
- The *MultiWoZ* dataset (Budzianowski et al., 2018) is a task-oriented dialogue corpus focused on service-oriented interactions (e.g., hotel booking, restaurant reservations, transportation inquiries), with no concept annotations.
- The *ESConv* dataset (Liu et al., 2021) is a specialized corpus for emotional support research. It contains 1,000+ multi-turn dialogues where users articulate personal struggles (e.g., workplace stress, interpersonal conflicts, self-esteem issues), and support providers respond with empathy, validation, and actionable guidance. It is annotated with 11 emotions and 8 strategies.
- The *EmpatheticDialogues* dataset (Rashkin et al., 2019) contains conversations centered on recognizing and responding to emotions (e.g., joy, sadness, frustration). It has annotations of user emotions but no strategies.

Dataset statistics. Table 9 summarizes the statistics of ESConv, DailyDialogue, and EmpatheticDialogues. Table 10 provides definitions of support strategies in ESConv and DailyDialogue.

B More Evaluation Details

Automatic Metrics. **Bleu-2(B-2)**(Papineni et al., 2002) first compute the geometric average of the modified n -gram precisions, p_n , using n -grams up to length N and positive weights w_n summing to one. Next, let c be the length of the prediction and r be the reference length. The BP and Bleu-2 are computed as follows.

$$BP = \begin{cases} 1 & \text{if } c > r \\ e^{(1-r/c)} & \text{if } c \leq r \end{cases}. \quad (3)$$

$$Bleu = BP \cdot \exp \left(\sum_{n=1}^N w_n \log p_n \right). \quad (4)$$

Rouge-L (R-L)(Lin, 2004) propose using LCS-based F-measure to estimate the similarity between two summaries X of length m and Y of length n , assuming X is a reference summary sentence and Y is a candidate summary sentence, as follows:

$$\begin{aligned} R_{lcs} &= \frac{LCS(X, Y)}{m} \\ P_{lcs} &= \frac{LCS(X, Y)}{n} \\ F_{lcs} &= \frac{(1 + \beta^2) R_{lcs} P_{lcs}}{R_{lcs} + \beta^2 P_{lcs}} \end{aligned} \quad (5)$$

Where $LCS(X, Y)$ is the length of a longest common subsequence of X and Y , and $\beta = P_{lcs}/R_{lcs}$ when $\partial F_{lcs}/\partial R_{lcs} = \partial F_{lcs}/\partial P_{lcs}$. In DUC, β is set to a very big number ($\rightarrow \infty$). Therefore, the LCS-based F-measure, i.e. Equation 5, is Rouge-L.

CIDEr (Vedantam et al., 2015) score for n -grams of length n is computed using the average cosine similarity between the candidate sentence and the reference sentences, which accounts for both precision and recall:

$$CIDEr_n(c_i, S_i) = \frac{1}{m} \sum_j \frac{\mathbf{g}^n(c_i) \cdot \mathbf{g}^n(s_{ij})}{\|\mathbf{g}^n(c_i)\| \|\mathbf{g}^n(s_{ij})\|}, \quad (6)$$

where $\mathbf{g}^n(c_i)$ is a vector formed by $g_k(c_i)$ corresponding to all n -grams of length n and $\|\mathbf{g}^n(c_i)\|$ is the magnitude of the vector $\mathbf{g}^n(c_i)$. Similarly for $\mathbf{g}^n(s_{ij})$.

Higher order (longer) n -grams are used to capture grammatical properties as well as richer semantics. (Vedantam et al., 2015) combine the scores from n -grams of varying lengths as follows:

$$CIDEr(c_i, S_i) = \sum_{n=1}^N w_n CIDEr_n(c_i, S_i), \quad (7)$$

Empirically, Vedantam et al.(Vedantam et al., 2015) found that uniform weights $w_n = 1/N$ work the best. So, We also use $N = 4$.

LLM-as-a-Judge. Table 11 provides the detailed prompt we feed to GPT4o for comparison evaluation.

Table 9: Statistics of ESConv, DailyDialogue, EmpatheticDialogues.

Category		ESconv	DailyDialogue	EmpatheticDialogues
# Sessions		1.3K	13.1k	2.5K
# Utterances		38K	103.0k	11.0K
Average # Utterances		28.9	7.9	4.3
Average Utterance Length		18.8	13.6	16.7
Seeker/Speaker1	# Utterances	20K	53.8k	5.7K
	Avg # Utterances	15.4	4.1	2.2
	Avg Uttr Len	16.8	13.2	20.8
	# Strategies	-	4	-
	# Emotions	11	7	32
Supporter/Speaker2	# Utterances	18K	49.2k	5.2K
	Avg # Utterances	13.6	3.9	2.1
	Avg Uttr Len	21.0	14.1	12.3
	# Strategies	8	4	-
	# Emotions	-	7	32

Table 10: Strategy definitions and abbreviations of ESConv and DailyDialogue.

Dataset	Strategies	Abbreviation	Definitions
ESconv	Question	Que.	Inquiring about problem-related information to help the seeker clarify their issues, using open-ended questions for best results and closed questions for specific details.
	Restatement or Paraphrasing	Res.& Par.	A simple, more concise rephrasing of the help-seeker’s statements that could help them see their situation more clearly.
	Reflection of Feelings	Ref.	Articulate and describe the help-seeker’s feelings.
	Self-disclosure	Self-Dis.	Divulge similar experiences that you have had or emotions that you share with the help-seeker to express your empathy.
	Affirmation and Reassurance	Aff. & Rea.	Affirm the help seeker’s strengths, motivation, and capabilities and provide reassurance and encouragement.
	Providing Suggestions	Pro.	Provide suggestions about how to change, but be careful to not overstep and tell them what to do.
	Information	Inf.	Provide useful information to the help-seeker, for example with data, facts, opinions, resources, or by answering questions.
	Others	others	Exchange pleasantries and use other support strategies that do not fall into the above categories.
DailyDialogue	Inform	-	Provide factual or contextual information that the speaker believes the listener may not know or is unaware of.
	Question	-	Seek specific information from the listener, assuming they possess the knowledge being requested.
	Directives	-	Express the speaker’s intention for the listener to take an action, including requests, instructions, or suggestions.
	Commissive	-	Indicate the speaker’s commitment to perform certain actions, such as accepting or rejecting requests or offers.

Human evaluation criteria. We enlisted four interns as human evaluators to rate the models across multiple dimensions, specifically Fluency, Emotion, and Satisfaction. The Satisfaction dimension encompasses four sub-aspects: Acceptance, Effectiveness, Sensitivity, and Alignment. Throughout this evaluation, we strictly adhered to international regulations and ethical norms, ensuring that all procedures comply fully with the guidelines regarding participant participation and data integrity. Each evaluator was required to assess every sample independently and strictly in accordance with the pre - pre-established criteria. By upholding these principles, the evaluation process maintained objectivity, standardization, and consistency, thereby improving the overall quality and credibility of the evaluation results.

The detailed manual scoring criteria are as follows:

- Fluency:

- 1: The sentence is highly incoherent, making it extremely difficult to understand and failing to convey a meaningful idea.
- 2: The sentence has significant incoherence issues, with only parts of it making sense and struggling to form a complete thought.
- 3: The sentence contains some incoherence and occasional errors, but can still convey the general meaning to a certain extent.
- 4: The sentence is mostly fluent with only minor errors or slight awkwardness in expression, and effectively communicates the intended meaning.
- 5: Perfect. The sentence is completely fluent, free of any errors in grammar, punctuation, or expression, and clearly conveys the idea.

Table 11: The Comparison Evaluation Prompt Content by GPT-4o from (Madani et al., 2024)

Prompt	
Please act as an impartial judge and evaluate the quality of the responses provided by two AI assistants to the user questions. You should choose the assistant that follows the provided strategy more carefully and precisely to answer the user's last utterance. You should use your emotional support expertise and knowledge to judge the quality of the response considering how well the answer follows the provided strategy. Your evaluation most importantly should consider strategy adherence and then the overall quality, naturalness, consistency and coherence of the final utterance. Begin your evaluation by comparing the responses of the two assistants and provide a short explanation. Avoid any position biases and ensure that the order in which the responses were presented does not influence your decision. Do not allow the length of the responses to influence your evaluation. Do not favor certain names of the assistants. Be as objective as possible. After providing your explanation, output your final verdict by strictly following this format: "JUDGE: [[A]]" if assistant A is better, "JUDGE: [[B]]" if assistant B is better, and "JUDGE: [[C]]" for a tie. Conversation history: {conversation_history} </The Start of Assistant A's Response> {assistant_a_resp} </The End of Assistant A's Response> </The Start of Assistant B's Response> {assistant_b_resp} </The End of Assistant B's Response>	
• Emotion: 1: The emotional expression is extremely inappropriate and chaotic, not in line with the content, and may convey wrong emotions. 2: The emotional expression has obvious flaws, either too weak or exaggerated, and is disjointed from the content. 3: The emotional expression is average. It can convey basic emotions but lacks depth and has minor issues. 4: The emotional expression is good. It can effectively convey the intended emotion with an appropriate intensity and is well integrated with the content. 5: The emotional expression is excellent. It is rich, nuanced, and perfectly matches the content, capable of evoking a strong and appropriate emotional response. • Acceptance: 1: The response inescapably triggers emotional resistance. 2: The response is highly likely to trigger emotional resistance. 3: The response has a possibility of emotional resistance occurring.	4: The response rarely provokes emotional resistance. 5: The response has no occurrence of emotional resistance. • Effectiveness: 1: The response actually worsens the seeker's emotional distress. 2: The response carries the risk of increasing stress levels, and this outcome varies depending on the individual user. 3: The response fails to alter the seeker's current emotional intensity and keeps it at the same level. 4: The response shows promise in calming the emotional intensity; however, it is overly complicated or ambiguous for the user to fully comprehend and utilize effectively. 5: The response appears to be highly effective in soothing the seeker's emotions and offers valuable and practical emotional support. • Sensitivity: 1: The response renders inaccurate evaluations regarding the seeker's state. 2: The response is characterized by rash judgments, as it lacks adequate assessment and in-depth exploration of the seeker's state.

3: The response is formulated with a one-sided judgment and a limited exploration of the seeker’s state.

4: The response demonstrates an understanding that only covers a part of the seeker’s state.

5: The response precisely grasps the seeker’s state and is appropriately tailored according to the seeker’s actual situation.

- Alignment:

1: The response is in total contradiction to the predicted strategy.

2: The response has a minor deviation from the predicted strategy.

3: There is some ambiguity between the response and the predicted strategy.

4: The response largely matches the predicted strategy, yet it contains some ambiguous elements.

5: The response effectively makes itself consistent with the predicted strategy.

- Satisfaction:

1: The response is extremely disappointing. It doesn’t answer the question at all and is of no help.

2: The response is poor. It only gives a partial answer and leaves many doubts unresolved.

3: The response is average. It meets the basic requirements but isn’t particularly outstanding.

4: The response is good. It answers the question clearly and provides some useful details.

5: The response is excellent. It not only answers the question perfectly but also offers valuable additional insights.

At the end, the cross-validation is conducted with the averaged results collected.

C Detailed Introductions of Baselines

We introduce the baselines with more details:

- Direct: directly inference the LLM.
- Direct-Refine: a straightforward refinement method in which the model revises its initial response to incorporate emotional support considerations.

- Self-Refine (Madaan et al., 2023): generates feedback about the emotional support based upon its initial response, then refines its response based on this feedback.

- ECoT (Li et al., 2024): an analogy of Chain-To-Thought prompt (Wei et al., 2022) on the emotional conversation, which first generates the seeker’s *emotion*, which then guides the generation of strategy and response.

- SoT (Ning et al., 2024): first guides LLMs to generate the skeleton of the answer, and then in parallel decodes to complete the contents.

- ToT (Yao et al., 2023a): decomposes complex problems into intermediate thought steps (like a tree structure), where LLMs evaluate the validity of each step and iteratively explore multiple reasoning paths to select the optimal solution, enhancing the ability to handle multi-step logical tasks.

- Plan-and-Solve (Wang et al., 2023): first prompts LLMs to generate a detailed plan outlining sub-goals and reasoning strategies, then executes the plan step-by-step to complete the solution, integrating planning and execution to improve the coherence and completeness of responses, especially in mathematical reasoning and multi-turn decision scenarios.

- RAG (Fan et al., 2024): the retrieval-augmented generation, which chooses the appropriate strategy by a semantic retriever, and includes it in the prompt to improve the response quality. In this work, we employ E5-Large (Wang et al., 2024) as the retriever.

- CSIM (Zhou et al., 2024): a self-chat method which produces the response based on its implicit simulation of future dialogue.

D More Experimental Results

D.1 More Typical Cases

Table 12 provides typical cases in DailyDialogue. From the contents generated, we can observe that CoCT can recognize the concept list, generate a chain of concepts, and then produce the detailed response based on the concept skeleton. The concept flow induces the LLM’s reasoning path. Table 13 shows cases of MultiWoz which show that CoCT adapts well to MultiWoz’s topics.

D.2 Bad Case Analysis

Table 14 provides a typical bad case. The red part indicates that CoCT attempts to transfer to the topic of ‘Hobbies and crafts’, however, which is almost the same as the original topic. Improvement of foundation models may alleviate such issues.

Table 12: Cases of CoCT in DailyDialogue, in contrast to Direct Inference.

Case 1	user	<i>I'm sorry I'm so late . I had a really bad day.</i>
	Direct	It's ten after six. We're late. But dinner is at six thirty.
	CoCT	<Commissive> No need to apologize; I'm just glad you're here now. <Question> Do you want to talk about your day? Sometimes sharing can really help.
Case 2	user	<i>But Sunday is the same with Saturday. If you don't mind, that's OK.</i>
	Direct	Hum.Maybe it will be better. The most important thing is that I can have a day to relax after a week's work .
	CoCT	<Surprise> Oh, really? I never noticed that. <Question> Are you sure Sunday is just as busy as Saturday? Maybe we could go early in the morning when it's less crowded?
Case 3	user	<i>Believe it or not, tea is the most popular beverage in the world after water.</i>
	Direct	Well, people from Asia to Europe all enjoy tea.
	CoCT	<Inform> Yes, that's true! Tea is indeed the most widely consumed beverage in the world after water. Its popularity spans across many cultures and countries, with different varieties like black, green, white, and oolong tea.

Table 13: Case of CoCT on MultiWOZ, in contrast to the original response.

user	<i>I will be needing a place to stay on the west side of town.</i>
assistant	Hobsons house is in the west part of town, would you like to stay there?
CoCT	<Affirmation and Reassurance> You've got a good head start on planning your trip to Cambridge. <travel> For accommodations on the west side. I can suggest checking out hotels near the River Cam or universities like Churchill College, what's your budget like for the stay?

Table 14: Bad case example of CoCT in ESconv.

user	<i>I just need ways to find validation that are not based on my job. But validation doesn't pay the bills.</i>
CoCT	<Reflection of feelings> It sounds like you're feeling a bit lost and uncertain about your sense of purpose and identity outside of your job. <Questioning> Can you tell me more about what you mean by "validation", what does that look like for you, and what are some things that have given you a sense of fulfillment and self-worth in the past? <Hobbies and crafts> Perhaps we can explore some hobbies or activities that bring you joy and help you feel more connected to yourself and others.